Judgment Gates for Paper Market Making: A Latency-Aware Pilot and Prospective Protocol

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Research draft; prospective results pending

Abstract

Across 26,525 reconstructed-book comparisons, we observe 0 mismatches, and construct 1,765 labelled development states from Binance and Hyperliquid BTC/ETH feeds. A logistic gate with isotonic calibration runs in 2.583 μs at the median and 18.135 μs at the empirical 99th percentile on Binance BTC, yet worsens Brier score on 3 of 4 markets; BTC changes from 0.2513 for rules to 0.3188. The original completed remote batches produce 0/134 valid answers within one second, including failures in the denominator. Successful GLM-5.3-flash judgments through OpenCode have median/99th-percentile client latency 7.159/21.935 s; NVIDIA-hosted GPT-OSS-20B yields 16.387/124.343 s. Jev inference is blocked by the gateway's paid-credit restriction and has no measured latency. We provide conservative queue replay, narrow probabilistic gates, coupling and framing protocols, and a frozen paper-only cohort. A correctness amendment leaves all 32 development replay scenarios unchanged. Sparse, dependent first-day observations do not support time-block confidence intervals, a calibrated remote combination, or a claim of improved market-making returns. The prospective test is registered separately and is ongoing; its final results are not evaluated. A separate descriptive forward audit reconciles 2,645,260 ledger records: the abstaining classifier submits no orders, and active variants lose after fees and inventory effects. This monitored P&L is not used for tuning or final inference. The bounded framing extension returns 54/74 valid judgments; only GPT-OSS and Nemotron Ultra supply paired results, with descriptive state-bootstrap intervals rather than population invariance evidence.

1 Introduction

A market maker needs both a pricing rule and a decision about whether the current state is suitable for quoting. A language model may provide contextual judgment, but its usefulness depends on the quality, calibration and arrival time of that judgment. A fast answer is not necessarily a useful trading signal; a probability is not calibrated merely because an API supplies it. An emotionally neutral interface also does not establish invariance to irrelevant framing.

We study four questions: whether route-level decision latency fits block or second cadence; whether a local classifier and a remote model improve each other; whether identical market features receive different answers under irrelevant loss, win, drawdown or clock context; and whether predicted probabilities match observed event frequencies. We keep deterministic pricing, inventory limits and risk checks in code. Models may estimate a typed probability or a coarse regime; they cannot generate prices or place orders.

Our contributions are an auditable first-day pilot with negative and missing results retained; event-driven replay that charges measured decision delays and excludes provably historical trades

from queue consumption; and a prospective protocol binding policy, source tag, continuity and ledger tail. We also provide self-contained rescoring from committed derived receipts. These are engineering and protocol contributions, not evidence of profitable trading or a completed comparison of all requested models.

2 Related work

Avellaneda and Stoikov (2008) formulate inventory-aware quoting in a limit order book. We use an explicitly simplified implementation as a common deterministic core. The information component of the spread in Glosten and Milgrom (1985) motivates measuring post-fill adverse moves, but our event label is a volatility proxy, not an identified estimate of informed trading.

The Brier score evaluates probabilistic forecasts (Brier, 1950). Zadrozny and Elkan (2002) study probability calibration from classifier scores, and Guo et al. (2017) illustrate that predictive confidence and calibration can diverge. Neither result guarantees that isotonic calibration helps on a small, changing crypto venue; that is measured here. Calendar-block uncertainty reflects the need to respect dependence rather than resample individual ticks (Kunsch, 1989). Multiple strategy attempts also matter: the deflated Sharpe ratio addresses selection and non-normal returns (Bailey and de Prado, 2014). We withhold its canonical value when the dispersion across all attempted policies is unavailable.

For Jev we use Vercel's typed Decision API, with the underlying service documented by TypeSafe (TypeSafe AI, n.d.; Vercel, n.d.). Documentation defines an interface, not an empirical latency or calibration result. No promotional article or social media post supplies evidence for this study.

3 Data

The pilot uses public Binance USDⓈ-M futures and Hyperliquid L2/trade streams for BTC and ETH, recorded on 2026-10-29 (UTC). An immutable local snapshot and its hashes are retained. The sibling recorder's `verify` compares reconstructed books with stored top-book observations: 26,525 comparisons pass with 0 errors. This validates internal reconstruction consistency, not exchange truth, queue position or 26,525 independent predictive observations.

Market	Labels	Train	Validation	Pilot	Isotonic
Binance BTC	609	294	142	153	98
Binance ETH	632	306	148	158	102
Hyperliquid BTC	301	144	71	76	48
Hyperliquid ETH	223	108	51	56	36

Table 1: Development labels and purged chronological split counts. The isotonic segment is inside the training period; counts need not sum across purged boundaries.

Features are spread in basis points, top-five depth imbalance, signed trailing trade flow, lagged mid return and trailing volatility. Feature computation follows receive/processing availability and uses only previously observed events. Let m_t be the observed mid. The target is

$$y_t = \mathbf{1}\left\{10^4 \left|\log(m_{t+10\mathrm{s}}/m_t)\right| > \max(1, \mathrm{spread}_{t,\mathrm{bp}}/2)\right\}.$$

The endpoint is the first available state at the horizon, with at most two seconds of tolerance. Development labels reject endpoints from a different session. Overlapping horizons and simultaneous

markets make these observations dependent. Hyperliquid's slower full-book updates are carried forward between arrivals; the sampling grid does not create independent exchange updates. Exact historical raw replay requires access to the retained private snapshot. Calibration and latency were recomputed from derived receipts in the private research archive; the public export provides aggregate statistics and precomputed paper artifacts.

4 Method

4.1 Deterministic quoting and execution

For relative-return variance v_t, inventory q_t, unit order size s, risk aversion γ, horizon T and arrival parameter κ, the implemented reservation and width are

$$r_t = m_t[1 - (q_t/s)\gamma v_t T], \qquad w_t = m_t \left[\gamma v_t T + \frac{2}{\gamma}\log(1+\gamma/\kappa)\right].$$

The code rounds bids downward and asks upward using decimal tick arithmetic, checks post-only eligibility on arrival, and enforces frozen inventory limits. Parameters are assumptions shared by every gate, not maximum-likelihood estimates of the venue's arrival process. Maker/taker fees are assumed to be 2/5 basis points, respectively; no account fee schedule is claimed.

Quotes join behind displayed queue. Trades with the correct aggressor side first consume that queue, then remaining quote size. Optimistic replay permits touching trades; pessimistic replay requires strict traded-through prices as well as queue consumption. Cancellation does not improve queue position. Quotes outside known finite depth are withheld. These scenarios are sensitivity models, not proven upper and lower bounds on actual exchange fills.

Pending cancel/replace keeps old quotes exposed until the measured judgment delay expires; it cannot be superseded by a newer pending answer. A state older than the registered age limit triggers a hold. Separate exchange streams can deliver old trades after a quote has arrived. Each quote therefore records the maximum exchange timestamp already observed at entry, from either book or trade stream. Trades at or before this bound, or lacking an ordering timestamp, cannot consume queue or fill. Only timestamps within the same venue clock are compared. The bound excludes demonstrably old trades; exact exchange order-entry time between observations remains unobserved.

4.2 Gates, coupling and abstention

Rules and a standardised logistic classifier use identical numeric features. Logistic regularisation is fixed; isotonic calibration uses a later, purged part of training. Expanding chronological folds provide development diagnostics. Abstention thresholds minimise validation Brier loss with a fixed rejection cost; rejected states do not submit quotes. A zero-profit, zero-coverage policy is not interpreted as a superior market maker.

Remote clients use the same market features for narrow typed judgments. Jev asks a Boolean veto through `/v1/evaluate`; its probability is the shared toxicity-proxy target. It uses no chat messages; thresholds, prices and risk remain in code. The older TypeSafe-compatible access probe is retained as historical evidence. A fresh cheap decision preflight precedes the resumable calibration, framing and coupling command. The registered targets include Jev, GLM-5.3-flash, DeepSeek-v4.1-flash and GPT-OSS, with a large model restricted to minute-cadence regime judgments. Client time includes transport, queueing, retries and, for OpenCode, CLI startup; shared NIM-slot waiting is

excluded. Retries and failures are retained. Cached response reuse is not a new physical latency measurement. Historical aliases do not reveal immutable served model revisions.

A validation-fitted stack forms $p = wp_C + (1 - w)p_R$. An uncertainty cascade calls the remote gate only inside a validation-selected classifier uncertainty band; its replay charges both local and remote delay on those decisions. A combined confidence threshold permits abstention. Neither fitted weights nor financial outcomes are reported without a complete common validation/pilot prediction matrix. The large-model regime mask also honours the arrival time of each update.

4.3 Calibration and framing

We report Brier $= n^{-1} \sum_i (p_i - y_i)^2$, equal-width reliability bins, and ECE $= \sum_b (n_b/n)|\bar{p}_b - \bar{y}_b|$. ECE depends on binning and sample size; isotonic levels can be few. Framing tests keep market features fixed and vary only irrelevant recent P&L, drawdown wording or clock context. A neutral repeat measures generation variability. We measure signed/absolute probability shifts and decision flips. Local numeric gates are deterministic, with no prompt state; this is an implementation property, not psychological evidence. Alongside the UTC-block analysis, a review-requested descriptive bootstrap resamples paired market-state differences (2,000 draws, seed 42, percentile 95% intervals, at least two pairs). These post-hoc development intervals assume exchangeable pairs, have no multiplicity adjustment and do not establish independent-day precision. Neutral repeats retain generation variability.

5 Protocol

The split is chronological: the first half trains, the next quarter validates, and the last quarter is an exploratory pilot. Full label horizons are purged at boundaries. Calibration uses a purged internal training segment, with scaling fit on training only. Repeated experiments on this pilot remain development; they are not repeated held-out tests. No seed is selected for a favourable outcome.

The ongoing prospective local cohort is `forward-v4`, from 2026-10-30 inclusive to 2027-01-28 exclusive in UTC. Its 32 engines span rules/classifier, with/without abstention, and both fill scenarios across all markets. Numerical coefficients, cutoffs and limits remain fixed. Earlier commissioning tags were withdrawn for decimal queue lookup, finite-depth handling and finally ordering/ scoring defects; their histories and losses remain available. The review amendment leaves 32/32 development executions unchanged. Remote gates are excluded from this local cohort and need a separate future registration if access and validation predictions become adequate.

The primary family contains the pessimistic, non-abstaining classifier-minus-rules 10-second markout comparison in each market. We compare equal-weight matched-hour means, not unrelated individual fills. UTC hour/day block resampling uses 2,000 replicates and seed 42. The primary family uses Bonferroni $\alpha = 0.05/4$; optimistic comparisons and other intervals are labelled descriptive. Fewer than five blocks yields no CI. The one-basis-point half-width target additionally requires five complete days with at least 90% decision coverage. Coverage enumerates missing engines and boundary hours. Annualised risk ratios require a pre-window P&L baseline, boundary observations and 90% coverage in every registered hour, as well as adequate variable hourly returns; missing hours are never compressed.

The frozen pre-test register counts 698 variants/iterations conservatively, including correlated attempts. All-trial Sharpe dispersion is unavailable. Canonical DSR is therefore null; retained-variant sensitivity is explicitly separate and is also not estimable in this short pilot. No all-trial DSR significance claim is made.

Prospective decisions and fills carry continuity identities. Explicit gaps, process starts and session changes censor pending labels and markouts; horizon-end and endpoint-tolerance censoring are reported separately. A label never bridges a known outage. The final scorer checks policy bytes against its immutable Git tag, validates every start identity and verifies the complete hash chain through a captured checkpoint tail. Scoring is reserved by cohort identity, independently of output filename. This assumes the cohort directory and local anchors are trusted; it does not defeat an attacker replacing all anchors or copying the cohort to bypass a local reservation. No final test has been evaluated. A separate bounded framing extension fixes 8 named NIM models and at most 28 judgments per model on four evenly spaced original development states and all framing contexts. It uses the shared two-slot mutex, sequential requests, a 30-second read timeout, no timeout retries, backoff on rate/server errors and a three-failure cutoff. It neither tunes a financial policy nor adds rows to the frozen prospective cohort.

6 Results

6.1 Calibration: a fast local gate can be worse

Market	n	Brier R	Brier C	ECE R	ECE C
Binance BTC	153	0.2513	0.3188	0.2482	0.3371
Binance ETH	158	0.2520	0.3802	0.0945	0.3631
Hyperliquid BTC	76	0.1791	0.1671	0.1405	0.0908
Hyperliquid ETH	56	0.2436	0.3347	0.1914	0.3123

Table 2: Exploratory local calibration on identical per-market pilot labels. R denotes rules; C denotes logistic plus isotonic calibration. Lower is better.

Table 2 reports all markets. The classifier worsens Brier on 3 of 4 markets. Its Binance BTC score rises from 0.2513 to 0.3188 despite very fast inference. Figure 1 shows bin-level behaviour; neither sparsely occupied bins nor a few isotonic levels justify an assertion of reliable confidence. The pilot is too short for the registered block CIs.

Market	n	Δ Brier (C–R)	UTC hours	Block 95% CI
Binance BTC	153	0.0675	1	–
Binance ETH	158	0.1283	1	–
Hyperliquid BTC	76	-0.0120	1	–
Hyperliquid ETH	56	0.0911	1	–

Table 3: Paired classifier-minus-rules Brier differences on existing development labels. Positive is worse for the classifier. Each market occupies one UTC hour; the minimum independent-block requirement is unmet, so intervals remain absent.

Table 3 retains paired differences rather than treating overlapping labels as independent observations. Across 96 development markout cases (all gates, fill scenarios and horizons), 0 support a day-block CI, with at most 1 UTC day represented. No one-basis-point test precision claim follows. More bootstrap draws cannot replace observed days; the prospective final scorer remains reserved until its registered end.

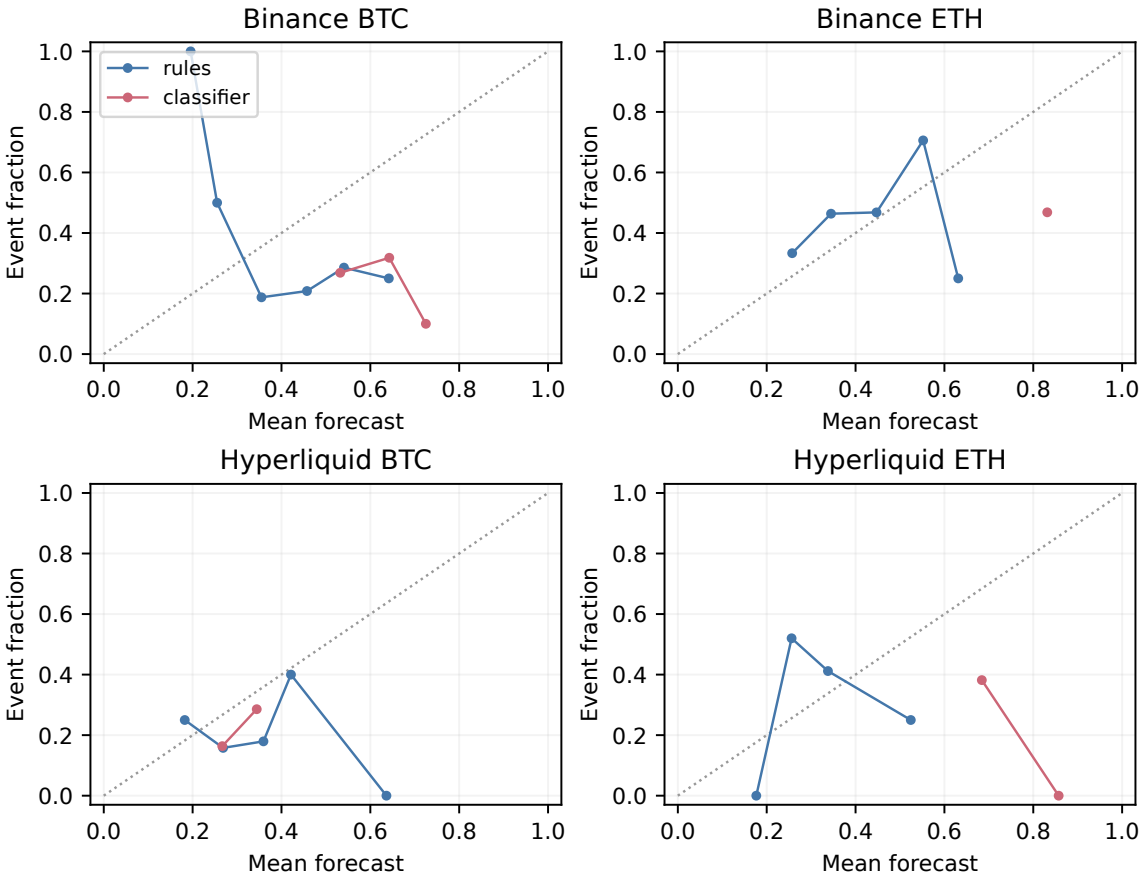


Figure 1: Reliability diagrams generated from committed pilot receipts. Lines join occupied bins for readability; they are not fitted calibration functions. There are no confidence bands because the independent-block requirement is unmet.

6.2 Latency: measured routes do not support subsecond judgment

Model / route	Valid	Failed	p50 (s)	p99 (s)
GLM-5.3-flash / CLI	20	36	7.159	21.935
DeepSeek-v4.1-flash / CLI	0	30	–	–
GLM-5.3-flash / NIM	0	3	–	–
DeepSeek-v4.1-flash / NIM	0	3	–	–
GPT-OSS-20B / NIM	26	5	16.387	124.343
Nemotron Ultra / NIM	5	4	32.265	51.029
Jev / Gateway	not run	access 403	–	–

Table 4: Original completed remote batches. Quantiles condition on valid answers and are unstable at these sample sizes. Failed judgments retain retries/timeouts. Jev access rejection is not an inference failure or latency sample.

The stored Binance BTC local benchmark gives 2.583/18.135 μs for classifier p50/p99. Table 4 reports the remote routes separately. The original batches and the explicit CLI quota recheck achieve 0/134 valid judgments within one second; all failures remain in the denominator. These are route/configuration-specific observations, not universal lower bounds on model speed. Figure 2 includes failures in its denominator rather than plotting only successful latency CDFs. It excludes the separate quota recheck, whose failures are counted in the deadline receipt, and does not pool configurations.

NIM flash judgments time out in the tested configuration; this does not establish that the models themselves cannot answer quickly. The original Nemotron run has both valid answers and truncated reasoning outputs. A later budget configuration returns HTTP 400 on every attempted request and cannot supply a regime replay. Jev's documented gateway route returns an account paid-credit restriction despite positive promotional credit; no native key exists. We measure neither native Jev nor gateway inference latency. Gateway rejection time is not substituted for it. Cost per million operational decisions is unmeasured: free/trial access and token catalogue prices cannot establish production unit cost, failure cost or coverage.

6.3 Execution: sparse fills and no established economic gain

Fees, inventory and adverse mid moves can dominate a small captured spread. Tables 5–6 make low activity and missing markouts visible. A post-fill markout overlaps inventory P&L and is not added a second time to the cash decomposition. Optimistic results, hypothetical taker flattening and complete fill/equity records remain in the receipts. The pilot cannot support annualised Sharpe/Sortino or a deflated performance conclusion. It has not shown that a faster gate, classifier, or abstention improves economically meaningful market-making outcomes.

6.4 Provisional forward activity and fee drag

The read-only audit verifies the full checkpoint hash chain and reconstructs fill cash and inventory independently before reconciling each engine's latest mark. Table 7 never sums cumulative balances across days. The abstaining classifier submits 0 orders and has 0 quote-present observations across 257,584 decisions per fill scenario. Its zero P&L is inactivity. The no-abstention variants retain toxicity vetoes, stale-state holds and hard risk limits; they are not unconditional quoting policies.

In the pessimistic scenario, rules and classifier capture respectively 0.562 and 0.778 basis points of filled notional, below the assumed 2.000 basis point maker fee. Spread capture thus fails to cover

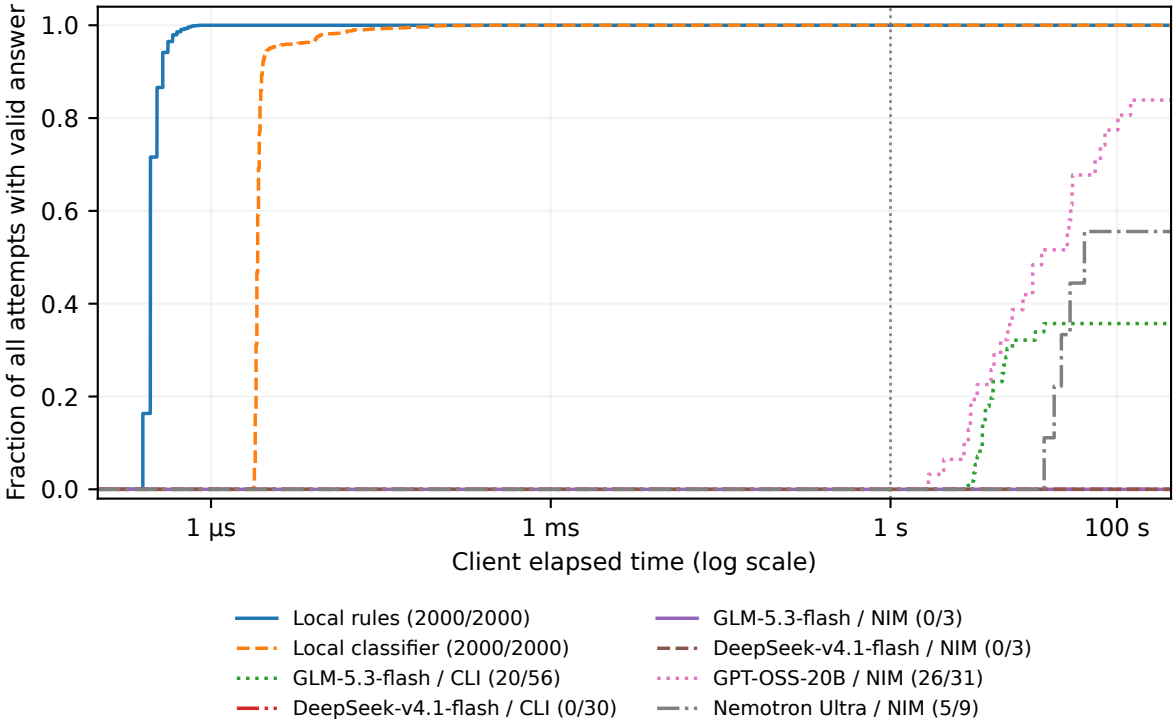


Figure 2: Valid-answer arrival fractions in the original batches; legend counts are valid/all attempts. NIM resource-slot waiting is excluded; CLI startup and request retries are included. Zero-success curves overlap at zero. Jev has no inference curve. No confidence band is supported.

Market / gate	Fills	Spread ($)	Inventory ($)	Fees ($)	P&L ($)
Binance BTC / R	4	0.00370	-0.08860	0.06415	-0.14905
Binance BTC / R+A	1	0.00805	-0.02570	0.01604	-0.03369
Binance BTC / C	0	0.00000	0.00000	0.00000	0.00000
Binance BTC / C+A	0	0.00000	0.00000	0.00000	0.00000
Binance ETH / R	1	0.00085	0.00300	0.00464	-0.00079
Binance ETH / R+A	0	0.00000	0.00000	0.00000	0.00000
Binance ETH / C	0	0.00000	0.00000	0.00000	0.00000
Binance ETH / C+A	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid BTC / R	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid BTC / R+A	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid BTC / C	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid BTC / C+A	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid ETH / R	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid ETH / R+A	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid ETH / C	0	0.00000	0.00000	0.00000	0.00000
Hyperliquid ETH / C+A	0	0.00000	0.00000	0.00000	0.00000

Table 5: All pessimistic development scenarios, in USD. R/C denote rules/classifier; +A adds validation-selected abstention. Inventory is the marked inventory component; spread plus inventory minus fees reconciles P&L. Zero fills are retained.

Market / gate	Fill rate	1 s bp (n)	10 s bp (n)	60 s bp (n)
Binance BTC / R	0.0208	-0.158 (3)	-0.249 (3)	-0.349 (1)
Binance BTC / C	0.0000	– (0)	– (0)	– (0)
Binance ETH / R	0.0036	-1.034 (1)	– (0)	– (0)
Binance ETH / C	0.0000	– (0)	– (0)	– (0)
Hyperliquid BTC / R	0.0000	– (0)	– (0)	– (0)
Hyperliquid BTC / C	0.0000	– (0)	– (0)	– (0)
Hyperliquid ETH / R	0.0000	– (0)	– (0)	– (0)
Hyperliquid ETH / C	0.0000	– (0)	– (0)	– (0)

Table 6: Pessimistic non-abstaining development diagnostics. Markout cells show mean signed mid move in bp and usable endpoint count; missing endpoints are censored. Fill rate is filled volume/submitted volume. No block CI is supported.

Gate / scenario	Fills	Spread ($)	Fees ($)	Inventory ($)	P&L ($)
C+A / opt.	0	0.00	0.00	0.00	0.00
C+A / pess.	0	0.00	0.00	0.00	0.00
C / opt.	7165	32.44	82.58	-145.54	-195.68
C / pess.	6714	30.47	78.31	-143.00	-190.84
R+A / opt.	693	3.15	9.58	-20.46	-26.88
R+A / pess.	663	2.94	9.19	-24.31	-30.56
R / opt.	19330	53.56	182.55	-236.12	-365.11
R / pess.	18257	49.43	175.81	-230.37	-356.75

Table 7: Descriptive cumulative paper monitoring through 2026-10-30 13:25:10 UTC, aggregated across markets from commissioning, including time before the test window. R/C denote rules/classifier, +A denotes abstention. The audited prefix contains 2,645,260 records. Both fill scenarios are retained; these are not final test results or confidence estimates.

fees even before inventory losses. The inventory column is the residual that reconciles marked P&L to fill-mid spread capture minus fees; markouts overlap it and must not be added again. The frozen pricing core lacks a fee-aware minimum spread, so the losses do not isolate gate quality. Monitoring does not trigger policy retuning or an early held-out significance claim.

6.5 Framing and remote calibration remain underpowered

Model / route	n_D	Drawdown [95% CI]	n_R	Repeat [95% CI]
GLM-5.3-flash / CLI	2	0.440 [-0.020, 0.900]	2	0.015 [0.010, 0.020]
GPT-OSS-20B / NIM	4	-0.327 [-0.795, 0.141]	4	-0.217 [-0.727, 0.075]
Nemotron Ultra / NIM	1	0.130 [–]	1	0.000 [–]

Table 8: Paired signed probability shifts relative to neutral framing. Neutral repeat is a separate fresh generation on the same features. Brackets show descriptive paired-state bootstrap intervals, not UTC-block intervals; small n is not evidence of invariance. A dash denotes fewer than two state pairs.

Model / route	n	Brier	ECE
GLM-5.3-flash / CLI	3	0.0063	0.0733
GPT-OSS-20B / NIM	4	0.5813	0.6455
Nemotron Ultra / NIM	1	0.9604	0.9800

Table 9: Sparse neutral remote calibration diagnostics. These use different tiny state subsets from the full local pilot and cannot rank models against each other.

Table 8 shows that irrelevant drawdown wording can coincide with large probability shifts, while a neutral repeat can also shift. The sample cannot reliably separate framing susceptibility from stochastic generation. Local rules and classifier are invariant because distractors never enter their numeric features. Jev and the unavailable flash route have no paired inference result in the original batches. Table 9 is a diagnostic rather than evidence of transferable calibration.

NIM model	Valid/tried	n_D	Drawdown [95% CI]	n_R	Repeat [95% CI]
GLM-5.3-flash	0/3	0	– [–]	0	– [–]
GLM-5.3	0/3	0	– [–]	0	– [–]
DeepSeek-v4.1-flash	0/3	0	– [–]	0	– [–]
Gemma-4-31B-it	0/3	0	– [–]	0	– [–]
MiniMax-M3.1-flash	0/3	0	– [–]	0	– [–]
Nemotron Lightning	0/3	0	– [–]	0	– [–]
Nemotron Ultra	27/28	3	0.0167 [-0.0300, 0.0800]	3	-0.0100 [-0.0500, 0.0200]
GPT-OSS-20B	27/28	4	0.2328 [0.0000, 0.4655]	4	0.0625 [-0.0050, 0.1300]

Table 10: Bounded review framing extension, separate from the original batches. Valid/tried counts judgments; n_D, n_R count complete drawdown/neutral-repeat pairs. Brackets are descriptive paired-state bootstrap intervals; missing pairs are retained. No independent-day precision or model ranking is established.

The extension returns 54/74 valid judgments. Table 10 includes every prespecified model, including failed or incomplete routes; it does not pool configurations into the original latency headline.

Only GPT-OSS-20B and Nemotron Ultra produce valid answers. Each has one timeout; GLM-5.3 returns HTTP 410, Nemotron Lightning returns unparseable answers, and the remaining routes time out. Failed judgments have no probability or framing interval and are not treated as invariant responses.

NIM model	Context	n	Shift [95% CI]	Flip fraction [95% CI]
Nemotron Ultra	loss	3	0.0100 [0.0000, 0.0300]	0.0000 [0.0000, 0.0000]
Nemotron Ultra	win	3	0.0133 [0.0100, 0.0200]	0.0000 [0.0000, 0.0000]
Nemotron Ultra	drawdown	3	0.0167 [-0.0300, 0.0800]	0.0000 [0.0000, 0.0000]
Nemotron Ultra	morning	3	-0.1030 [-0.3290, 0.0200]	0.0000 [0.0000, 0.0000]
Nemotron Ultra	night	3	0.0100 [-0.0300, 0.0700]	0.0000 [0.0000, 0.0000]
Nemotron Ultra	neutral repeat	3	-0.0100 [-0.0500, 0.0200]	0.0000 [0.0000, 0.0000]
GPT-OSS-20B	loss	4	0.1859 [-0.0003, 0.5580]	0.2500 [0.0000, 0.7500]
GPT-OSS-20B	win	4	0.0996 [-0.0004, 0.2992]	0.0000 [0.0000, 0.0000]
GPT-OSS-20B	drawdown	4	0.2328 [0.0000, 0.4655]	0.2500 [0.0000, 0.7500]
GPT-OSS-20B	morning	4	0.1190 [0.0000, 0.3570]	0.0000 [0.0000, 0.0000]
GPT-OSS-20B	night	3	0.1069 [0.0000, 0.3200]	0.0000 [0.0000, 0.0000]
GPT-OSS-20B	neutral repeat	4	0.0625 [-0.0050, 0.1300]	0.0000 [0.0000, 0.0000]

Table 11: Every non-neutral framing condition in the completed extension for models with valid pairs. Shifts are signed probability differences from neutral; flip fractions compare thresholded decisions. Brackets are unadjusted descriptive paired-state bootstrap intervals. Degenerate zero-flip intervals do not establish population invariance; independent time-block CIs remain unavailable.

GPT-OSS's drawdown interval touches zero; Nemotron Ultra's contains zero. Nemotron's small win-context shift has an interval excluding zero, but this is a tiny conditional cohort with no multiple-testing correction. Neutral repeats also vary. Table 11 retains all tested contexts, rather than selecting only the largest shift. These observations cannot establish an emotion-free gate or separate framing effects from generation variability reliably. Complete common predictions are absent, so stacks, cascades and remote abstention variants remain not run rather than being fitted to a convenient subset. Q1 has preliminary route measurements; Q2 remains untested; Q3 and Q4 have only limited development evidence. None has a well-powered population answer.

7 Limitations

The main limitation is observation time: overlapping samples from a first-day pilot do not establish precision across days or regimes. Small isotonic segments, a volatility-proxy label and repeated pilot inspection further limit generality. Providers, prompts, quotas, failure cutoffs and timing routes differ. Empirical p99 from a handful of successes does not estimate a reliable service-level tail; resource contention and client overhead can matter. Neither vendor latency nor vendor calibration claims are tested without valid Jev inference.

L2 aggregation hides individual queue priorities, hidden orders, cancellations, RPI liquidity and own market impact. Hypothetical fees omit funding and any account-specific discounts. Kill-on-gap is idealised: real cancellations could arrive late and unseen fills could occur. Source-time bounds leave uncertain ordering between observations. A laptop can suspend or disconnect despite launchd and idle-sleep inhibition; commissioning health proves neither uninterrupted operation nor future uptime. The full requested remote forward matrix and multi-month acceptance remain incomplete.

The private history includes hosted outputs and exchange-derived rows. Weight licences do not establish redistribution permission. The public export includes original code, synthetic tests, this paper and author-computed aggregate statistics; underlying market observations, provider responses and private history are excluded. There is no claim of a publicly available historical

raw archive or of an arXiv submission. These access limitations reduce independent empirical reproducibility.

8 Reproducibility

The frozen runtime commit is

`12ad479abb867f03503f02632ebf93a37a161538`

The policy and preregistration remain in the private research archive. Original experimental receipts record commands, seeds, configurations, producing commits, source hashes and available model revisions. The final mechanical replay was repeated without fitting or threshold changes; neither the public export nor its publication changes the frozen prospective policy.

Tables, numerical macros and figures were generated from committed private receipts, refusing uncommitted inputs. The public export includes those precomputed artifacts, `paper/generated/manifest.json` with published input/output hashes, and `EXPORT.json` with the source commit and exported file hashes. Public aggregate measurements are in `results/research-summary.json`, `results/framing-summary.json` and `results/forward-diagnostics.json`. These retain uncertainty and failure summaries while excluding individual states, provider answers, per-call records and simulated events. DOI/arXiv identifiers are in `paper/refs.bib`; the private archive retains resolver evidence.

```
uv sync --frozen
uv run pytest -q
tectonic -X compile paper/main.tex
```

These public commands run synthetic code tests and compile the supplied tables; they do not reconstruct historical fills or rescore provider calls. Those tasks require the retained private snapshot, transcripts and pinned recorder. A new public-stream recording is a new experiment. The final scorer refuses use before the registered end and reserves scoring by cohort. No exchange keys, accounts or order endpoints are used. Publication boundaries and official exchange/provider terms are documented in `RELEASE_POLICY.md`.

9 Conclusion

The pilot does not establish that low latency, typed probabilities or excluded emotional context improve market-making performance. The local classifier is fast but usually less calibrated than rules here; tested remote routes are slow or unavailable; framing and combination evidence is incomplete. The engineering response is a narrow gate around deterministic risk, explicit abstention and failure handling, and a separately frozen prospective test. A credible answer now requires future observed days and adequate model access, not more favourable tuning on this same pilot.

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